

Graph Attention-Based Topological Modeling for Diabetic Retinopathy Severity Prediction Using Fundus Pairs and Clinical Metadata

Model 2 Research Article | BRSET Diabetic Retinopathy Project

Abstract

This research article documents the second implemented architecture for diabetic retinopathy (DR) severity prediction. The model is inspired by graph neural network work on topological DR classification, but it is adapted into a patient-level multimodal pipeline. Each patient is represented using paired left-eye and right-eye fundus images plus selected clinical variables. A pretrained EfficientNet-B0 image encoder and tabular MLP produce patient embeddings, and a patient similarity graph is then classified using a Graph Attention Network (GAT). The work emphasizes an improvement trail rather than a single final run: full 5-class DR_ICDR grading, grouped 3-class severity modeling, partial oversampling, class-aware focal learning, graph-based relational reasoning, severe-case threshold calibration, and explainable AI (XAI). The strongest practical configuration maintained 87.80% test accuracy while improving balanced accuracy from 0.5921 to 0.6302 and severe DR recall from 0.5769 to 0.7692 after threshold calibration. XAI was added through Grad-CAM for fundus regions, SHAP for clinical metadata, and GAT attention for influential patient-neighbor reasoning.

Keywords: diabetic retinopathy, BRSET, Graph Attention Network, EfficientNet, multimodal learning, explainable AI, Grad-CAM, SHAP, patient similarity graph.

1. Introduction

Diabetic retinopathy is a progressive retinal complication of diabetes, and reliable automated screening requires both strong classification performance and clinically meaningful interpretation. The project proposal frames the problem as automated detection and prediction from retinal images and associated clinical data. The literature review also identifies a gap in single-modality methods, limited XAI adoption, and inconsistent handling of class imbalance across DR severity grades. This study responds to these gaps by building a multimodal graph-based architecture that combines visual retinal evidence, clinical metadata, and patient-to-patient relational structure.

Unlike a plain image classifier, the implemented model treats each patient as a node in a graph. Node features are built from both eyes and tabular metadata, while edges connect similar patients. This makes the model closer to a topological reasoning pipeline, where the prediction is affected not only by an individual patient representation but also by the learned influence of clinically or visually similar neighbors.

2. Relation to the Model 2 Reference Architecture

Sundar and Sumathy proposed diabetic retinopathy disease-level classification using topological features and graph neural networks. Their work motivates the use of graph structure for DR staging, where retinal features are converted into a relational representation and then classified by a graph model. The implementation in this project does not copy the original pipeline directly. Instead, it adapts the graph idea to BRSET by constructing a patient-level graph from multimodal embeddings and using attention-based graph reasoning.



Figure 1. Conceptual graph-based DR pipeline inspired by the topological GNN literature.

3. Dataset and Preprocessing Trials

The experiment uses the BRSET-style split with train, validation, and test CSV files and processed fundus images. The raw image-level split contained 13,015 training images, 1,632 validation images, and 1,619 test images. Images were paired into patient-level records by matching left and right eye images for the same patient, giving 6,094 training patients, 764 validation patients, and 754 test patients. Patient overlap checks were zero across train-validation, train-test, and validation-test, which reduces leakage risk.



Sample right-eye fundus image used to illustrate the data modality.



Sample left-eye fundus image used to illustrate paired-eye modeling.

The tabular preprocessing retained nine clinically useful variables: patient_age, patient_sex, diabetes, optic_disc, vessels, macula, macular_edema, drusens, and myopic_fundus. Categorical eye and binary-like clinical columns were standardized into numeric format, missing patient age was imputed by the median, and non-modeling or leakage-prone variables were excluded from prediction.

Trial / Approach	Main Action	Reason	Observed Effect
Patient-level pairing	Combined left and right eye images for each patient	BRSET contains two eyes per patient	Created 6,094 / 764 / 754 patient records
Feature selection	Kept 9 tabular variables	Reduce leakage and noise while preserving clinical signal	Produced compact 9-feature tabular input
Full 5-class DR_ICDR	Classes 0-4 kept separately	Most detailed severity grading	High accuracy but weak minority-class recall
Grouped 3-class severity	No DR, mild/moderate, severe	Clinically clearer under severe imbalance	Improved macro and balanced metrics
Partial oversampling	Oversampled minority classes only in training	Reduce class imbalance without changing validation/test	Improved minority exposure during training
Severe threshold tuning	Adjusted severe prediction threshold to 0.35	Prioritize severe DR recall	Raised severe recall to 0.7692

4. Proposed Architecture

The final architecture consists of two stages. In Stage 1, both fundus images are passed through a shared EfficientNet-B0 image encoder, and clinical variables are passed through a tabular MLP. These features are concatenated into a patient embedding. In Stage 2, the patient embeddings become graph nodes. A patient similarity graph supplies edges, and a GAT learns attention weights over incoming neighbors before predicting DR severity.

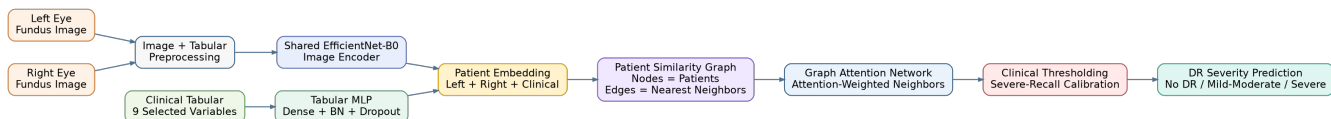


Figure 2. Implemented Model 2 architecture: multimodal patient embedding followed by graph attention reasoning.

5. Experimental Improvement Trail

The experiments were intentionally iterative. The first direction retained all five DR_ICDR classes, which preserves the most detailed clinical grading but suffers from extreme minority-class scarcity. The 5-class EfficientNet-tabular model reached high overall accuracy because class 0 dominates the dataset, but balanced accuracy and macro F1 remained low. Adding the GAT improved 5-class macro F1 from 0.1894 to 0.3184 and balanced accuracy from 0.1968 to 0.2957, showing that relational learning helped but did not fully solve severe imbalance.

The second direction grouped the task into three severity levels: No DR, Mild/Moderate DR, and Severe DR. This approach keeps clinically useful staging while reducing sparsity between adjacent classes. It also makes severe-case detection easier to optimize. Under this setup, the GAT achieved 0.8780 test accuracy, 0.5921 balanced accuracy, and 0.5472 macro F1. Severe-recall thresholding preserved accuracy but improved balanced accuracy to 0.6302 and macro F1 to 0.5526.

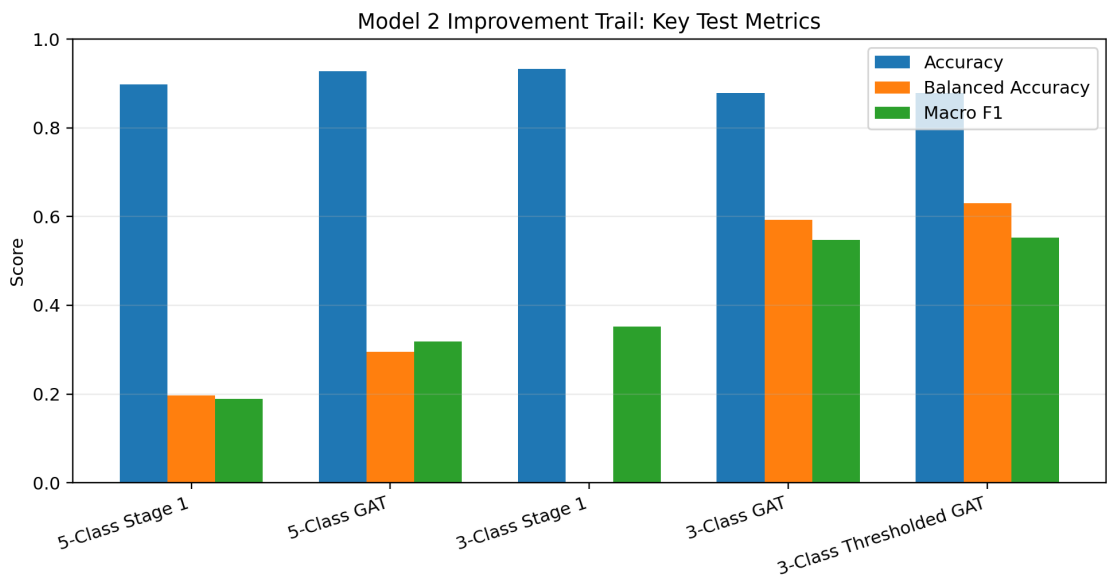


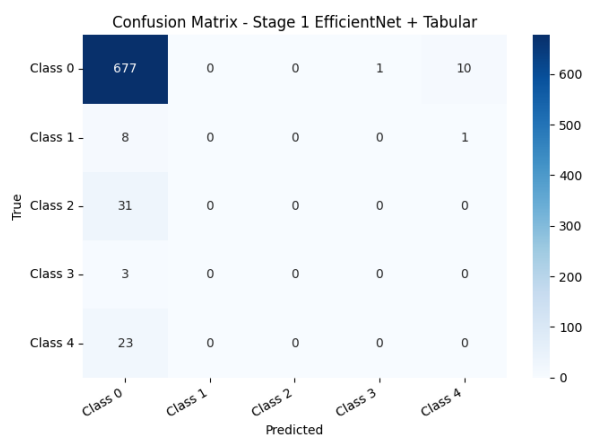
Figure 3. Key metric progression across the implemented improvement trail.

Approach	Accuracy	Balanced Acc.	Macro F1	Weighted F1	Important Finding
5-class EfficientNet + tabular	0.8979	0.1968	0.1894	0.8640	Collapsed mostly to class 0
5-class GAT	0.9271	0.2957	0.3184	0.8984	Improved minority handling but still class-imbalanced
3-class EfficientNet + tabular	0.9332	-	0.3521	-	High accuracy but weaker macro behavior
3-class GAT	0.8780	0.5921	0.5472	0.8893	Better severity-aware performance
3-class thresholded GAT	0.8780	0.6302	0.5526	0.8883	Severe recall increased to 0.7692

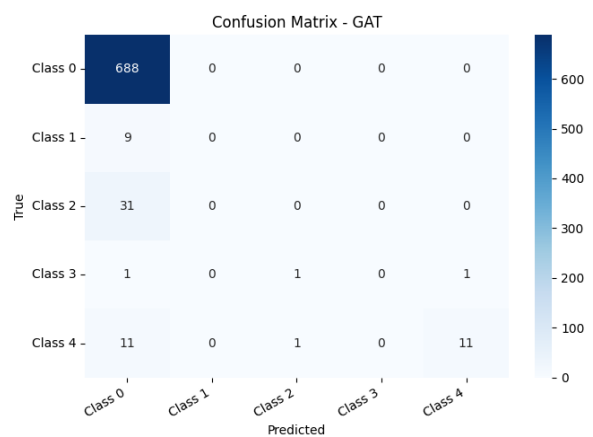
6. Quantitative Results

For the 3-class GAT test evaluation, the model produced the following class-level behavior: No DR precision 0.96 and recall 0.92; Mild/Moderate DR precision 0.20 and recall 0.28; Severe DR precision

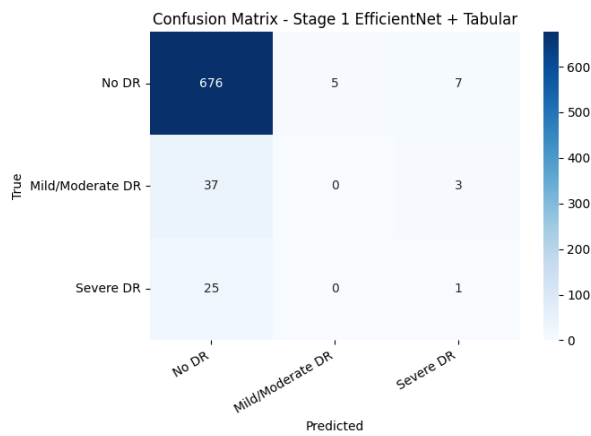
0.39 and recall 0.58. After threshold calibration, Severe DR recall increased to 0.77 and Severe DR F1 improved to 0.53, while overall accuracy stayed at 0.8780. This is a meaningful clinical trade-off because missing severe DR is more costly than slightly increasing false positives.



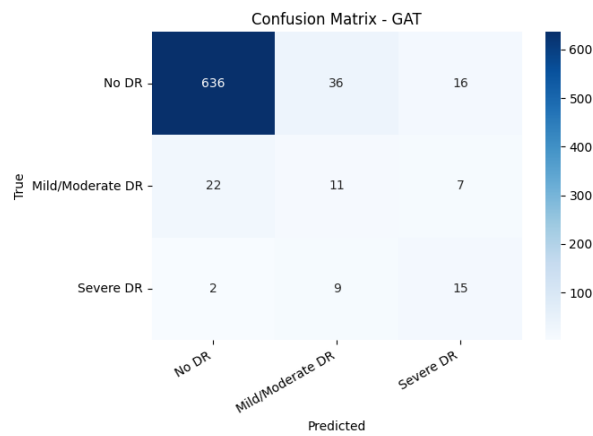
5-class Stage 1 confusion matrix.



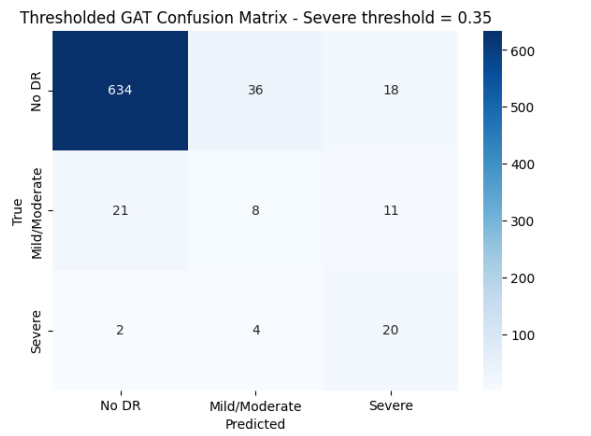
5-class GAT confusion matrix.



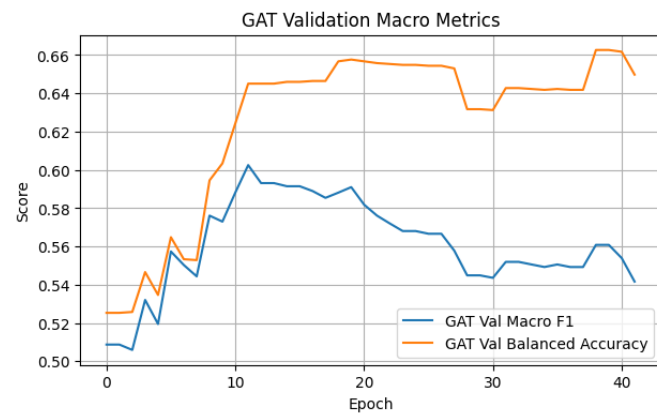
3-class Stage 1 confusion matrix.



3-class GAT confusion matrix.



3-class thresholded GAT confusion matrix.



3-class GAT validation macro metrics curve.

7. Explainable AI Techniques

The XAI design intentionally covers three explanation levels. Grad-CAM explains image evidence by highlighting retinal regions in the left and right eye fundus images. SHAP explains tabular contribution by estimating how clinical variables push the model toward or away from a predicted class. GAT attention explains graph reasoning by showing which patient-neighbor nodes received the highest influence during message passing. Together, these methods provide visual, clinical, and relational explanations rather than relying on a single post-hoc technique.

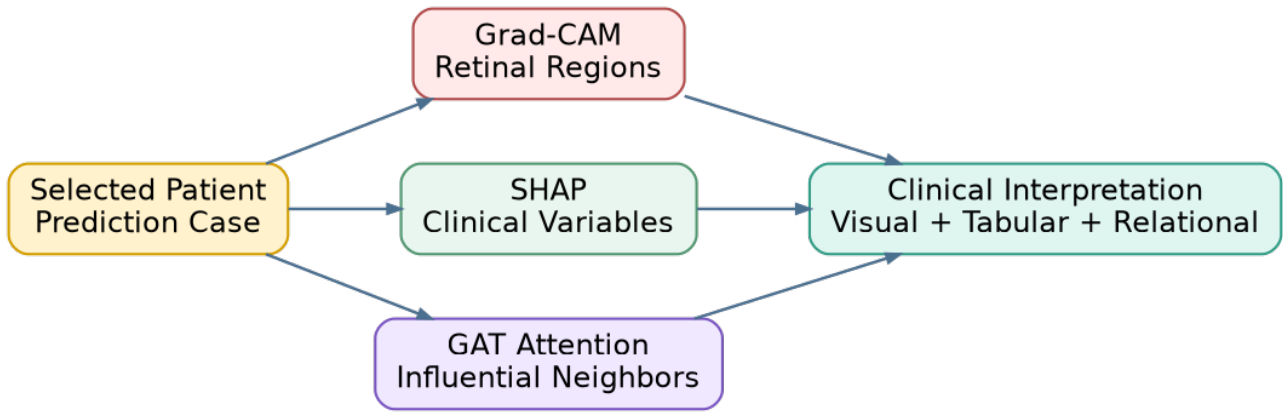
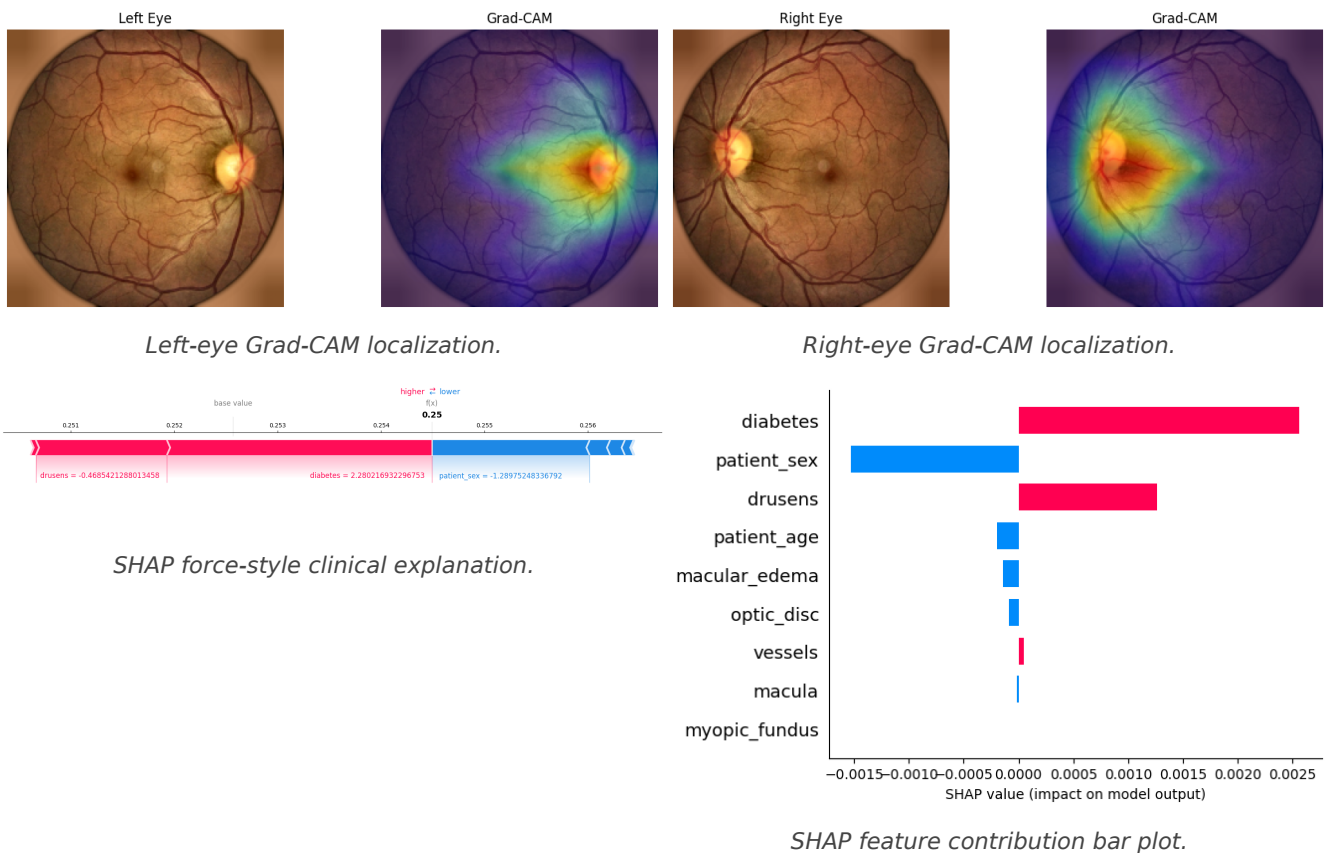


Figure 4. XAI workflow added to the graph-based model.



8. Discussion

The strongest result is not the highest raw accuracy; it is the improvement in severity-aware metrics after reformulating the task and calibrating the output. The 5-class task is clinically detailed but too sparse for several DR_ICDR levels in the available split. In contrast, the 3-class severity grouping better matches the amount of available minority data and produces more reliable macro performance. The GAT stage adds value by using patient-neighbor structure, and the severe threshold gives a controllable clinical operating point.

The main limitation is that minority-class performance is still substantially lower than No DR performance. This reflects the real distribution of the split and the difficulty of learning rare DR categories. Future work should test additional graph construction strategies, compare GAT against GCN and GraphSAGE, use calibrated probability methods, and validate on an external dataset to measure generalization. More clinically diverse severe examples would likely improve recall and precision simultaneously.

9. Conclusion

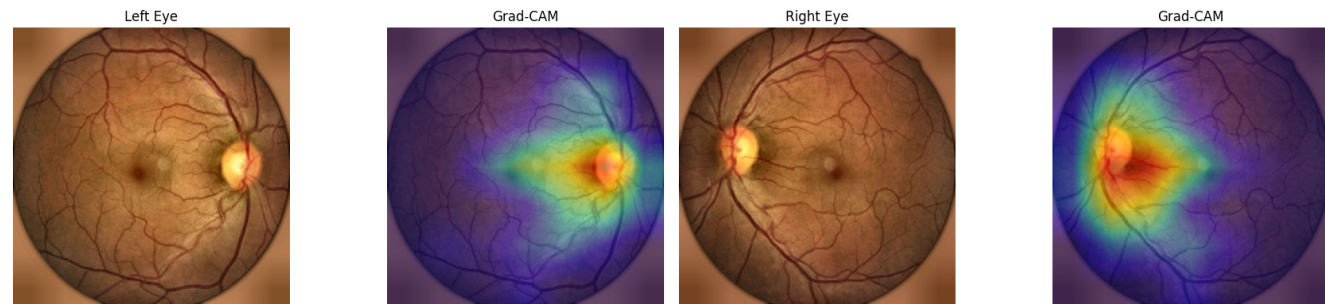
Model 2 demonstrates a professional graph-based DR severity pipeline that moves beyond single-image classification. The project improves from a highly imbalanced 5-class setup into a clinically more useful 3-class GAT system with severe-recall calibration and multi-level explainability. The final thresholded GAT maintains 87.80% accuracy, increases balanced accuracy to 0.6302, and raises severe DR recall to 0.7692. The architecture is suitable for research reporting because it clearly connects the literature-inspired graph concept to a reproducible multimodal implementation and interpretable outputs.

References

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Appendix A. Full Metric and Output Graphs

This appendix includes the metric plots, confusion matrices, training curves, validation curves, feature-map visualizations, and XAI outputs extracted from the implemented approaches. Duplicated or near-duplicated confusion matrices are kept because they came from separate evaluation stages.

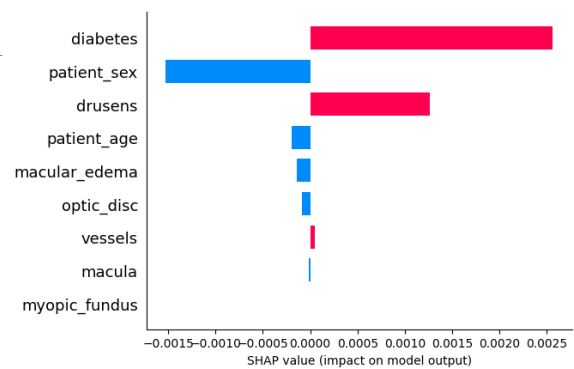


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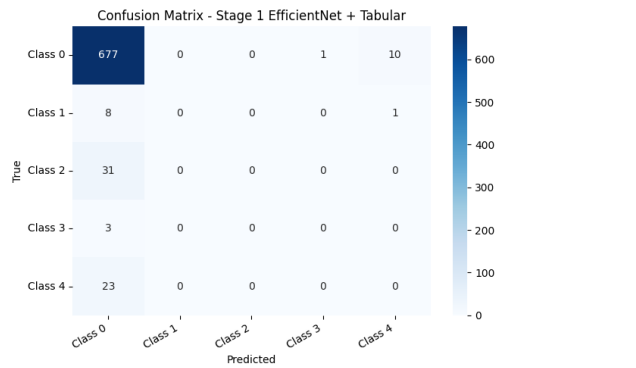
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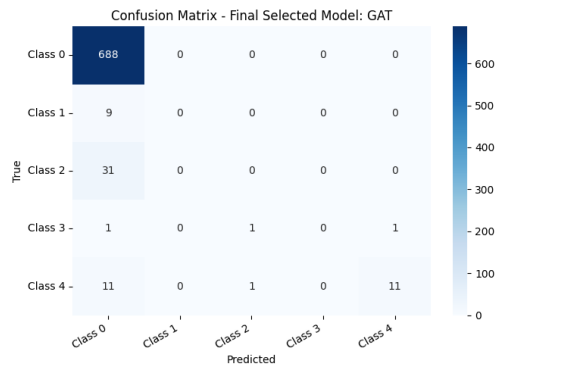
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5-class XAI cell03 img03

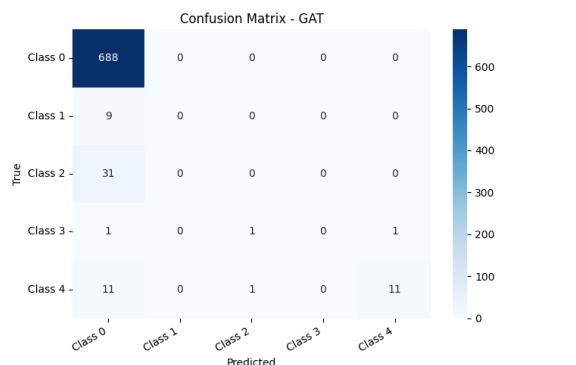


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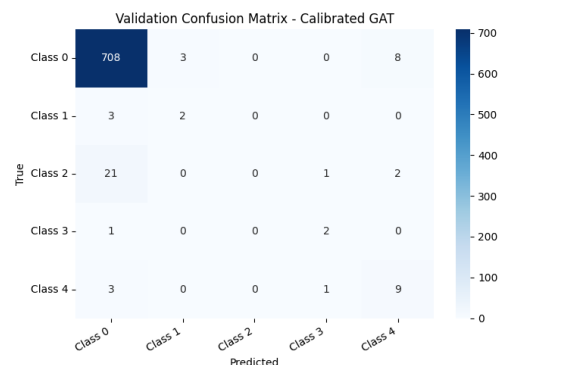


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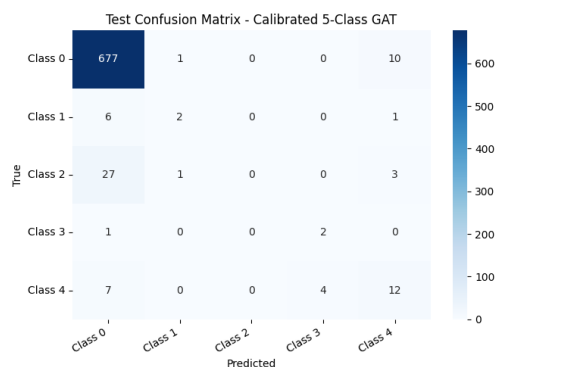
Appendix A. Full Metric and Output Graphs (continued)



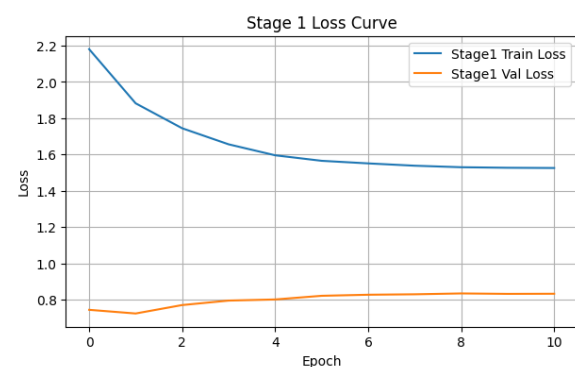
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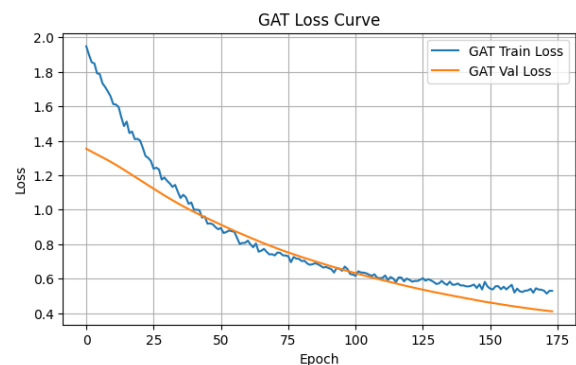
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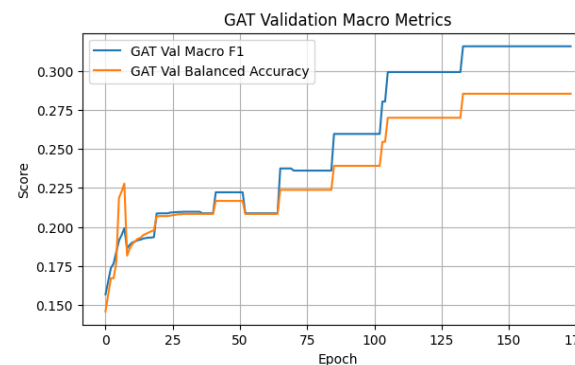
5-class cell35 img04



5-class cell36 img05

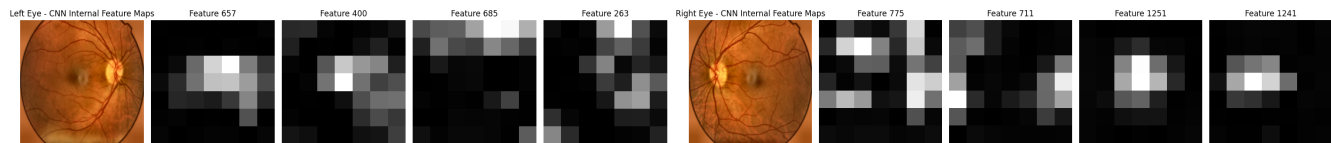


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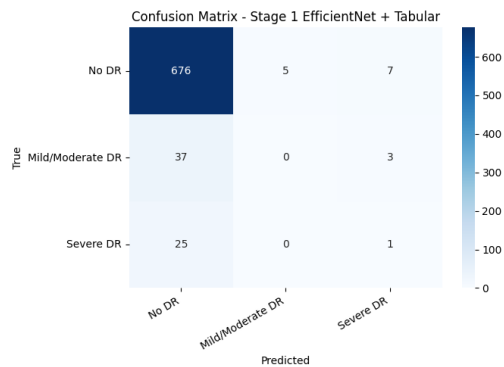


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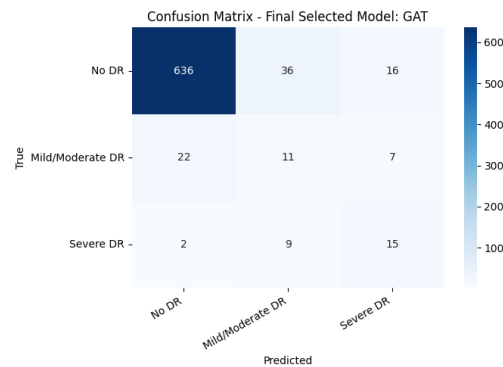
Appendix A. Full Metric and Output Graphs (continued)



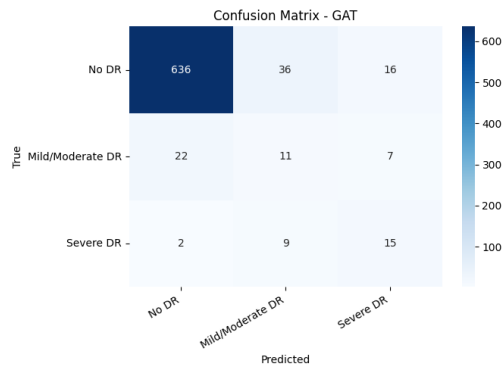
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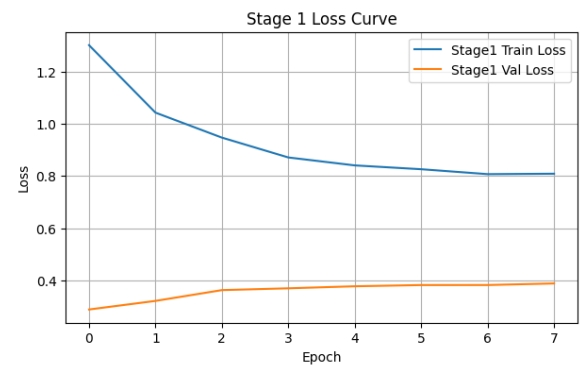
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3-class cell17 img00



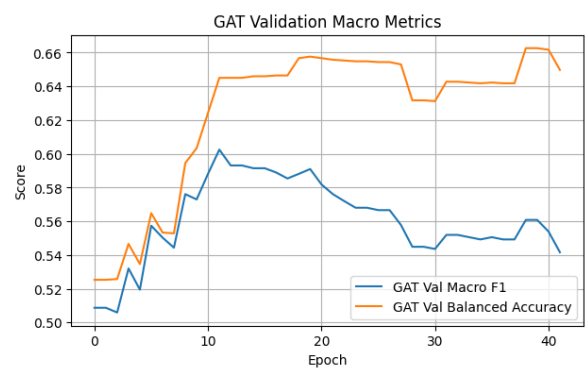
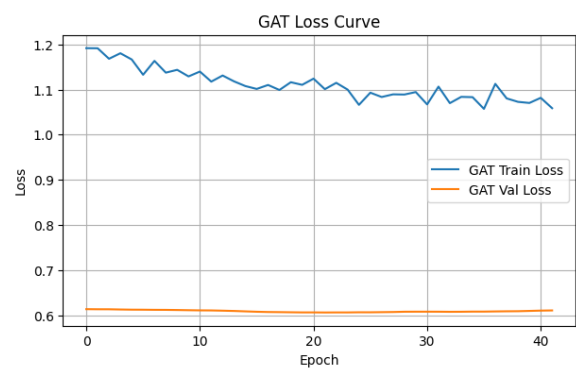
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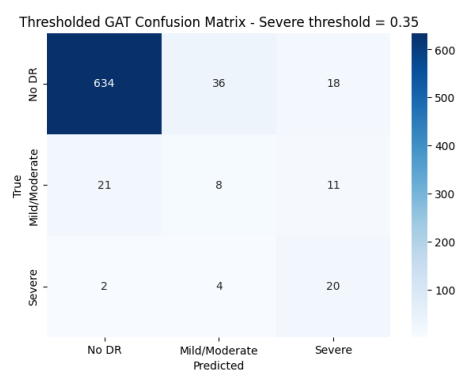
3-class cell32 img02

3-class cell33 img03

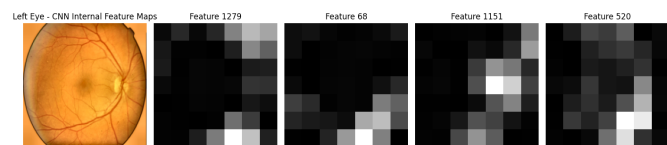
Appendix A. Full Metric and Output Graphs (continued)



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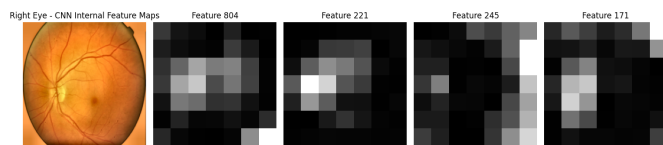


3-class cell33 img05



3-class cell40 img07

3-class cell37 img06



3-class cell40 img08